- **AVOID:** Avoid dehydration, vigorous exertion, and drugs that reduce preload (nitrates, diuretics).
- Advanced Tx: Septal reduction therapy. ICD for primary prevention if major risk factors exist (e.g., family history of SCD, massive LVH ≥30 mm, unexplained syncope).

3. **Restrictive CM: Treat the underlying cause**. Diuretics for symptom control. **Amyloidosis treatment:** Tafamidis.

PEARL: HCM ICD risk factors: family history of SCD, LVH ≥30 mm, unexplained syncope, NSVT, abnormal BP response to exercise.

Myocarditis

Keywords / Typical Presentation

- **Young patient + recent viral illness** (days to weeks prior) + chest pain, dyspnea, or new heart failure.
- May present like ACS (chest pain, troponin elevation, ST changes) but coronaries are clean.
- Can progress to dilated cardiomyopathy or cause sudden arrhythmic death.
- **Etiologies:** Viral most common (Coxsackie B, parvovirus B19, adenovirus, COVID-19). Also: Lyme (→ heart block), Chagas, giant cell (fulminant), sarcoidosis, drug hypersensitivity.

Diagnostic Pathway

- **Best test: Cardiac MRI** (myocardial edema + late gadolinium enhancement in non-coronary pattern).
- **Also obtain:** Troponin (elevated), EKG (diffuse ST changes/arrhythmias), echo (wall motion abnormalities, reduced EF). Coronary angiography if ACS suspected.
- **Endomyocardial biopsy:** Rarely needed; reserve for suspected giant cell myocarditis or rapidly deteriorating patient.

Core Principles for Myocarditis Management:

1. **Supportive care is the mainstay:** Treat heart failure per guidelines (ACEi, ARB, beta-blocker if stable, diuretics). Inotropes or MCS if cardiogenic shock.
2. **Activity restriction:** Avoid strenuous exercise for 3-6 months (risk of arrhythmia).
3. **Arrhythmia monitoring:** Patients at risk for VT or VF; may need wearable defibrillator (LifeVest) as bridge.
4. **Immunosuppression:** NOT routine. Reserved for biopsy-proven giant cell myocarditis, eosinophilic myocarditis, or cardiac sarcoidosis.
5. **Prognosis:** Most viral myocarditis resolves; ~30% progress to dilated cardiomyopathy. Giant cell myocarditis has high mortality without treatment.

PEARL: Athletes with myocarditis must abstain from competitive sports until inflammation resolves (including follow-up MRI, echo, Holter).